



Department of Computer Science ITU, Lahore
Pakistan

Instructions

Please submit a zip file titled as “**Bscs23xxx_A02.zip**”
You are required to solve the assignment in **C++**

Assignment Task:

Your objective is to implement a performance comparison between your custom vector class and the Standard Template Library (STL) vector. Specifically, you need to load 100 megabytes of data into both your custom vector implementation and the STL vector, measuring and comparing the time it takes for each to complete this operation.

Once you have successfully loaded the data, analyze and document the differences in performance between the two vectors. If one vector exhibits faster load times, provide a clear explanation as to why that might be the case.

Deliverables:

1. C++ Source Code:

Submit a single C++ file containing your full implementation for both vectors and the timing comparison.

2. README File:

The README should include:

- An overview of your custom vector implementation.
- A detailed summary of the results.
- Your observations and analysis of the performance difference, with a focus on why one vector may be faster than the other.